Transformers and ChatGPT KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

Word bank: 100 million 2017 attention Generative matter most next word
Nobel Prizes November 2022 Pre-trained Transformer

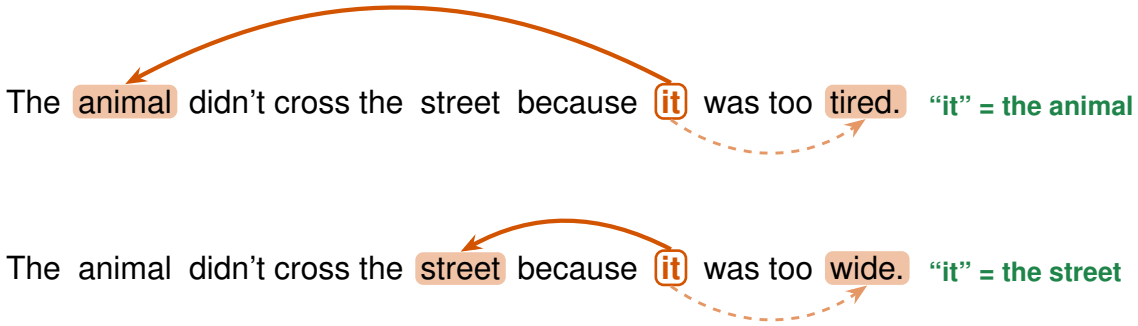
1. What does "it" mean?

The animal didn't cross the street because **it** was too **tired**. "it" means the animal

The animal didn't cross the street because **it** was too **wide**. "it" means the street

Which word told you? the last word: tired or wide

2. Attention.



A transformer looks at every word. For each word, it decides which other words matter most.

This is called attention. Google scientists shared the transformer in 2017.

3. GPT and ChatGPT.

G = Generative it makes new text

P = Pre-trained it learned first from a huge amount of text

T = Transformer it uses attention to connect the words

A chatbot writes by predicting the next word, again and again.

ChatGPT came out in November 2022. About 100 million people used it in about 2 months.

In 2024, two Nobel Prizes went to scientists who worked on AI.

4. Be the chatbot: next word, by vote.

Word	The word with the most votes	How many votes?
1	Answers vary.	
2		
3		
4		
5		
6		

Our class sentence: After school, my friends and I Answers vary.

The class AI tool wrote: Answers vary.

Was the class sentence **surprising** or **ordinary**? Why? Ordinary. The most common word wins.

5. Talk about it. ChatGPT learned from text that people wrote. Most writers were never asked.

Writers **should** / **should not** be asked first, because Answers vary.

Go Further (Day 2). The class timeline. Number the moments from 1 (first) to 6 (last).

4	ImageNet: 14 million labeled pictures	1	Alan Turing asks: Can a machine think?
6	ChatGPT	3	Deep Blue beats the world chess champion
2	ELIZA, one of the first chatbots	5	Transformers and attention

✂ *Cut out the cards. Add them to the class timeline.*

2017

Google scientists share the **transformer**. It uses attention.

Draw a picture here.

2022

ChatGPT comes out. About 100 million people use it in 2 months.

Draw a picture here.

2024

Scientists who worked on AI win two **Nobel Prizes**.

Draw a picture here.